



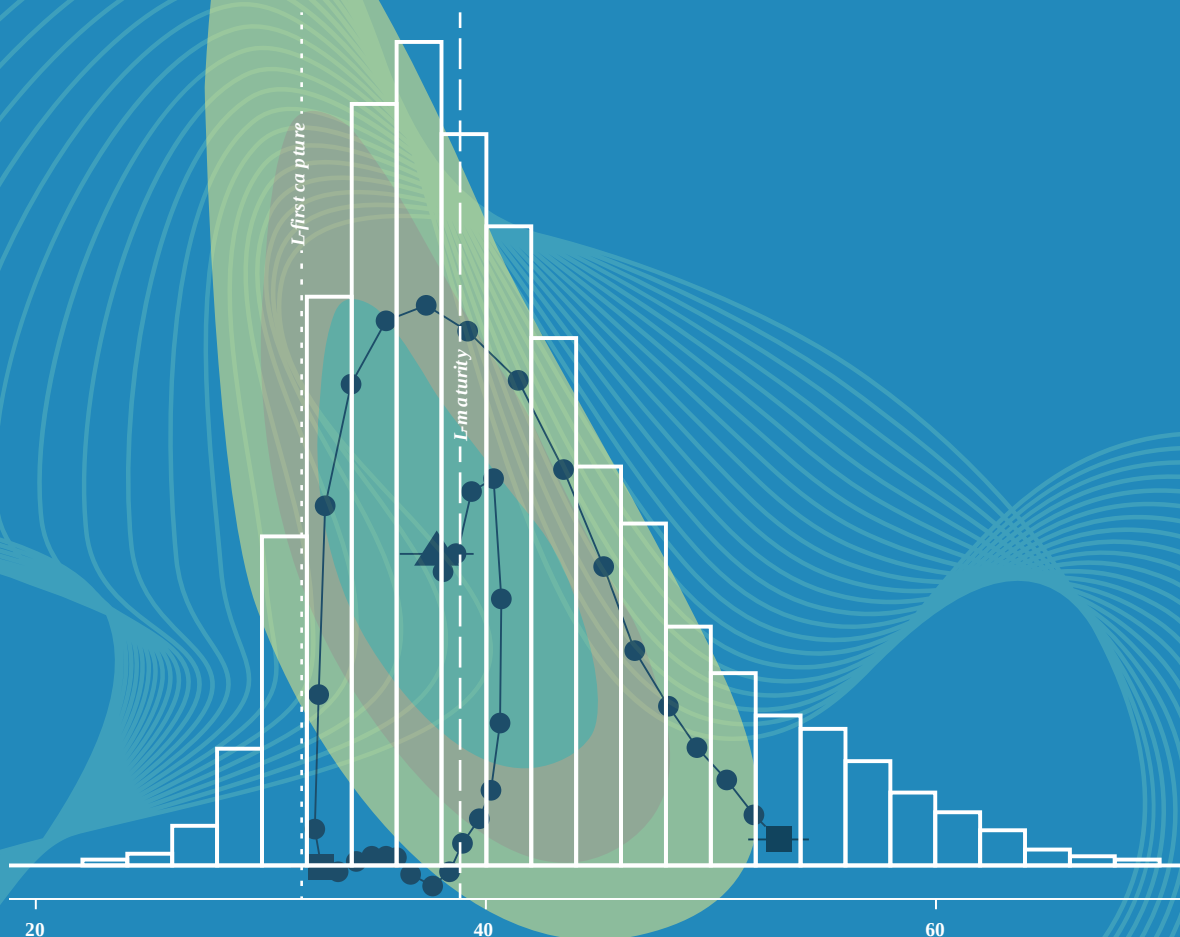
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Knowledge Management

Multi-stakeholder Knowledge Management Based
on Co-production Principles to Support
Fisheries Resource Management in Indonesia



Multi-stakeholder Knowledge Management Based on Co-production Principles to Support Fisheries Resource Management in Indonesia

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Irfan Yulianto

Chairperson, Rekam Nusantara Foundation

Knowledge management, as a formal concept, emerged in the late 20th century, born from the convergence of disciplines such as organizational theory, information science, and computer science. Although the idea of managing knowledge has roots in ancient practices—such as libraries—it gained prominence as modern organizations sought to adapt to the demands of the information age. Today, knowledge management stands as an inseparable tool that enabling organizations to systematically capture, organize, share, and leverage knowledge assets to drive better decision-making.

Rekam Nusantara Foundation is proud to present an overview of knowledge management tailored to the context of fisheries management. Indonesia, as one of the world’s largest maritime nations, has placed significant emphasis on fisheries management through a multi-stakeholder framework. This framework includes management authorities, scientific and consultative panels, all working together toward sustainable fisheries governance.

To achieve an ideal state of fisheries management, these diverse stakeholders must effectively collaborate within a robust knowledge management system. This book provides a detailed analysis of the roles, platforms, and interactions among stakeholders, to establish a unified and effective knowledge management platform that supports sustainable fisheries management in Indonesia.

We hope this technical recommendation escalate in the transformation systems of knowledge management platform, to support the collective efforts for a sustainable Indonesia’s invaluable marine resources. In addition, this book is intended to serve as an academic learning reference for university students and lecturer, by providing a structured understanding of knowledge management framework and their application in fisheries governance.

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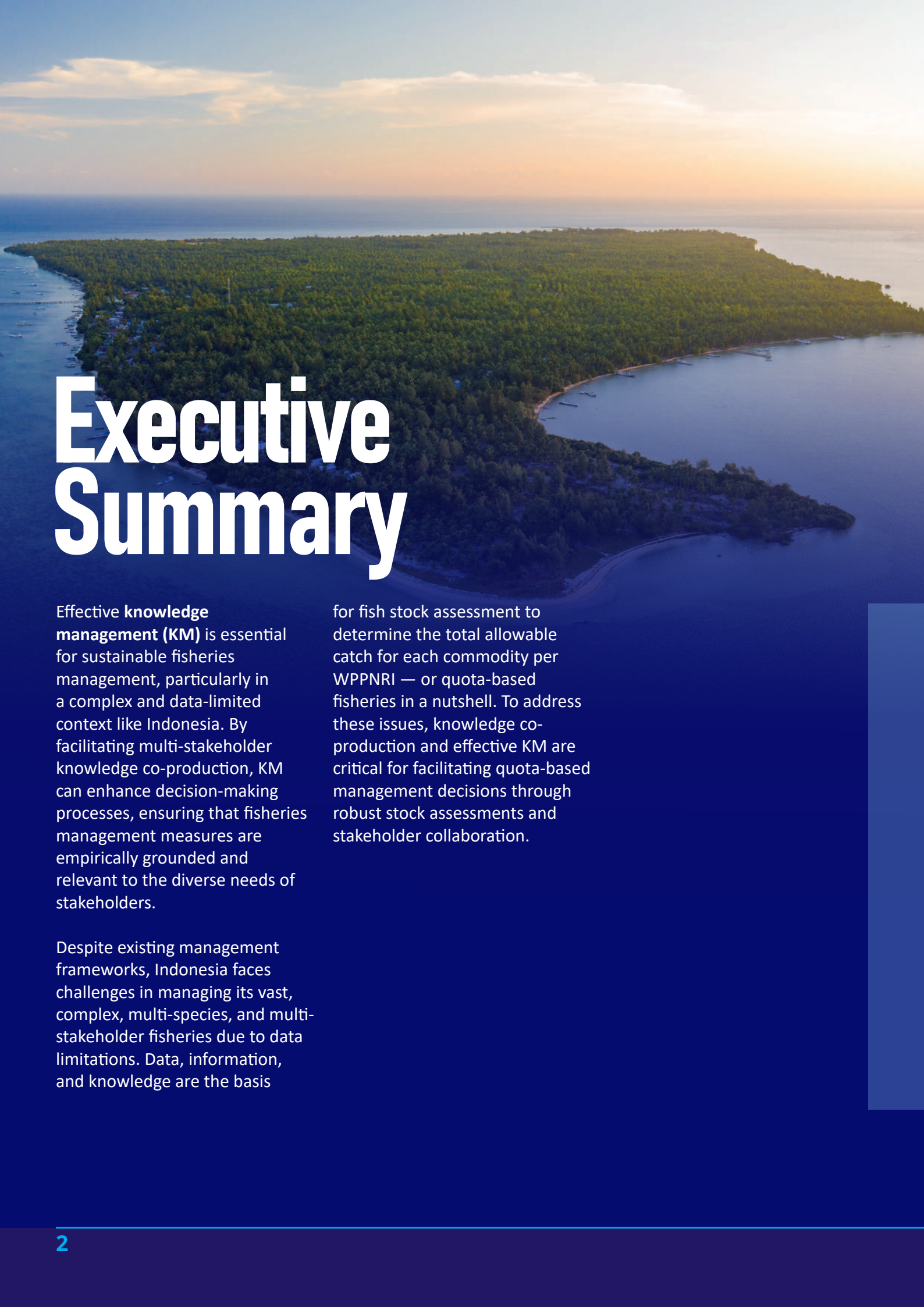
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Executive Summary

Effective **knowledge management (KM)** is essential for sustainable fisheries management, particularly in a complex and data-limited context like Indonesia. By facilitating multi-stakeholder knowledge co-production, KM can enhance decision-making processes, ensuring that fisheries management measures are empirically grounded and relevant to the diverse needs of stakeholders.

Despite existing management frameworks, Indonesia faces challenges in managing its vast, complex, multi-species, and multi-stakeholder fisheries due to data limitations. Data, information, and knowledge are the basis

for fish stock assessment to determine the total allowable catch for each commodity per WPPNRI — or quota-based fisheries in a nutshell. To address these issues, knowledge co-production and effective KM are critical for facilitating quota-based management decisions through robust stock assessments and stakeholder collaboration.

Based on our findings, we recommend establishing a Multi-Stakeholder Fisheries KM System.

An integrated digital KM system to streamline data collection, sharing, and coordination among stakeholders (e.g., scientific authorities, management bodies, NGOs, private sectors). Improved collaboration and accessibility will enable better fisheries management measures.

As a first step toward an integrated KM system, we recommend developing a **Fisheries Data Catalog**. A centralized data catalog will simplify data organization, making information retrieval faster and more effective. This initiative will also serve as a gateway to a more comprehensive KM framework.

We have identified key challenges and proposed solutions to each, namely:

1. Limited Funding

- **PROPOSED SOLUTION:** Explore alternative funding sources, including public-private partnerships and in-kind contributions.

2. Inadequate Infrastructure

- **PROPOSED SOLUTION:** Invest in technological upgrades, capacity-building programs, and stakeholder collaboration.

3. Human Resource Gaps

- **PROPOSED SOLUTION:** Promote outreach, establish specialized functional roles in government bodies, and leverage in-kind support.

4. Information Silos

- **PROPOSED SOLUTION:** Strengthen inter-stakeholder collaboration to foster better data and knowledge sharing.

The proposed KM initiatives aim to empower Indonesia's fisheries management system by fostering collaboration and evidence-based decision-making. The immediate establishment of a Fisheries Data Catalog will address urgent data management needs and lay the groundwork for a comprehensive, multi-stakeholder KM system. Addressing funding, infrastructure, human resource, and collaboration challenges is critical for successful implementation.

Using a stakeholder mapping exercise, based on power-interest and KM capability analyses, we identified the following stakeholders: Government (Ministry of Marine Affairs and Fisheries/ MMAF), the National Committee on Fish Stock Assessment (Komnaskajiskan), the National Research and Innovation Agency (BRIN), universities, Non-Government Organizations (NGOs), businesses/private sectors, and fishers.

- **Stakeholders with high KM capabilities:** Komnaskajiskan, BRIN, NGOs, and universities.
- **High-power, high-interest stakeholders:** Komnaskajiskan, MMAF, BRIN, and universities.
- **Underrepresented groups:** Fishers and private businesses exhibit high interest but have low power and KM capabilities, requiring targeted capacity-building efforts.

Ringkasan Eksekutif

Manajemen pengetahuan (Knowledge Management/KM) yang efektif sangat penting untuk menjaga kelestarian perikanan, terutama di Indonesia yang memiliki kondisi perikanan yang rumit dan seringkali kekurangan data. Dengan melibatkan berbagai pihak untuk berbagi dan menciptakan pengetahuan bersama (*co-production*), KM dapat membantu menghasilkan keputusan yang lebih baik. Hal tersebut memastikan bahwa pengelolaan perikanan didasarkan pada fakta dan sesuai dengan kebutuhan semua pihak yang terlibat.

Meskipun sudah ada kerangka kerja pengelolaan, keterbatasan data menyebabkan Indonesia menghadapi tantangan dalam

mengelola perikanannya yang luas, kompleks, terdiri dari berbagai spesies, dan melibatkan banyak pihak. Data, informasi, dan pengetahuan menjadi dasar untuk menghitung stok ikan dan menentukan total tangkapan yang diperbolehkan untuk setiap komoditas di setiap WPPNRI — atau dikenal sebagai penangkapan ikan terukur berbasis kuota. Untuk mengatasi tantangan ini, *co-production* dan KM yang efektif menjadi sangat diperlukan. Dengan cara ini, pengambilan keputusan dalam pengelolaan berbasis kuota berlandaskan penghitungan stok yang akurat dan terpercaya, dan melibatkan kolaborasi antar pemangku kepentingan.

Temuan

Kami menggunakan metode untuk memetakan berbagai pihak yang terlibat dalam pengelolaan perikanan, berdasarkan analisis relasi kewenangan-kepentingan dan kapabilitas KM. Dari hasil analisis ini, pihak-pihak utama yang teridentifikasi adalah Pemerintah (Kementerian Kelautan dan Perikanan/KKP), Komite Nasional Kajian Stok Ikan (Komnaskajiskan), Badan Riset dan Inovasi Nasional (BRIN), universitas, organisasi non-pemerintah (LSM), sektor bisnis/swasta, dan nelayan.

- **Pemangku kepentingan dengan kapabilitas KM tinggi:** Komnaskajiskan, BRIN, LSM, dan universitas.
- **Pemangku kepentingan dengan kewenangan dan kepentingan tinggi:** Komnaskajiskan, KKP, BRIN, dan universitas.
- **Kelompok yang kurang terwakili:** Nelayan dan sektor swasta menunjukkan kepentingan besar tetapi memiliki kewenangan dan kapabilitas KM yang rendah, sehingga mereka memerlukan kegiatan pengembangan kapasitas yang terarah.

Rekomendasi Utama: Membangun Sistem KM Perikanan Multi-pihak

Sistem KM digital yang terintegrasi akan memperlancar pengumpulan data, berbagi informasi, dan koordinasi antar pemangku kepentingan (misalnya, otoritas ilmiah, badan pengelola, LSM, sektor swasta). Kolaborasi dan aksesibilitas yang lebih baik akan mendukung program-program pengelolaan perikanan yang lebih efektif.

Sebagai langkah awal menuju sistem KM terintegrasi, kami merekomendasikan pengembangan Katalog Data Perikanan. Katalog data yang terpusat akan mempermudah penataan data sehingga pengambilan informasi menjadi lebih cepat dan efisien. Inisiatif ini juga dapat menjadi pintu masuk menuju kerangka KM yang lebih komprehensif.

Tantangan dan Solusi Utama:

1. Pendanaan Terbatas
 - **SOLUSI YANG DIUSULKAN:** Mengoptimalkan sumber pendanaan alternatif, termasuk kemitraan publik-swasta dan kontribusi in-kind.
2. Infrastruktur yang Tidak Memadai
 - **SOLUSI YANG DIUSULKAN:** Berinvestasi dalam peningkatan teknologi, program pengembangan kapasitas, dan kolaborasi pemangku kepentingan.
3. Kekurangan Sumber Daya Manusia
 - **SOLUSI YANG DIUSULKAN:** Mendorong kegiatan sosialisasi, membentuk peran fungsional khusus dalam badan pemerintah, dan memanfaatkan dukungan non-moneter.
4. Silo Informasi
 - **SOLUSI YANG DIUSULKAN:** Memperkuat kolaborasi antar pemangku kepentingan untuk meningkatkan berbagi data dan pengetahuan.

Inisiatif KM yang diusulkan bertujuan untuk memperkuat sistem pengelolaan perikanan di Indonesia dengan mendorong kerja sama dan pengambilan keputusan berbasis ilmiah. Pembuatan Katalog Data Perikanan menjadi salah satu langkah pertama untuk mengatasi kebutuhan mendesak dalam pengelolaan data dan menjadi dasar membangun sistem KM multi-pihak secara lebih menyeluruh. Agar inisiatif ini berhasil, beberapa tantangan perlu diatasi, seperti pendanaan, infrastruktur, tenaga ahli, dan kerja sama antara berbagai pihak.

Background

Effective knowledge management (KM) is crucial to support sustainable fisheries management. Inter-agency (multi-stakeholder) KM can enhance the capabilities of managers to make more relevant and empirically based decisions (Cooke et al., 2021). Additionally, structured and accessible knowledge enables fisheries managers to make more accurate decisions based on precise data and information. This is vital to maintain the balance of marine ecosystems, ensure the survival of various fish species, and support the livelihoods of communities dependent on the fisheries sector. Good KM at local and regional levels also supports policy and adaptation to climate change (Lindegren & Brander, 2018).

One of the main aspects of KM is data management. Data management involves collecting, storing and analyzing relevant fisheries data, such as fish stock data, catch results, and marine ecosystem conditions (Linke & Bruckmeier, 2015). Data management is important

for stakeholders to develop policies using fisheries data (Ribera-Altimir et al., 2023). Information gathered through data management can be integrated and analyzed to identify trends and patterns that can aid in decision-making. Poor data management can lead to uncertainties in fish stocks, potentially threatening the economy and food security of those dependent on fish stocks and increasing the risk of overfishing (Bradley et al., 2019).

Furthermore, to support KM and data management, the availability of fisheries data is a critical instrument in achieving sustainable fisheries. In reality, there are several challenges with fisheries data, such as data shortages due to inaccurate recording and reporting by fisheries operators. This also causes the options provided by the government to support fisheries management to not necessarily align with the actual conditions and issues in the field (Ramadhani et al., 2022). The diversity of agencies and institutions also

creates the potential for data overlap, as data collection and processing are conducted according to the needs of each work program. This often occurs due to a lack of awareness about data ownership among parties holding the data. This situation creates a gap in data availability, with crucial data sometimes not identified. Addressing data gaps and overlaps is essential for improving data availability. Cooperation among various data-collecting parties is needed to achieve this. There should be a design of an integrated data and information catalog so that data availability, data needs, and data stewards can be known in more detail and, subsequently, a framework for data collection to policy formulation in the context of marine and fisheries management can be developed.

In this paper, we discuss the current state of fisheries KM in Indonesia and identify the key stakeholders involved in fisheries resource management through exercising papers and interviewing stakeholders and experts. This includes examining each stakeholder's power, interest, and capacity in conducting KM, highlighting the importance of a KM framework that facilitates knowledge co-production to address issues related to limited and dispersed data. We examined the stakeholders' KM by reviewing two

scientific pieces of literature (Baskoro & Nugroho, 2011; Parikenan et al., 2020), 4 stakeholder's websites or news reports ^{1, 2, 3, 4}, and 3 government reports or regulations ^{5, 6, 7}. We also provide recommendations for developing a multi-stakeholder KM system.

¹ <https://www.brin.go.id/page/6/profil>

² <https://www.siej.or.id/index.php/id/ekuatorial/mdpi-dan-brin-perkuat-pengelolaan-perikanan-berkelanjutan-di-indonesia>

³ <https://perikanan.org/public/index.php/id/kegiatan/frci-brin-ipb-university-dan-ykan-beri-pelatihan-pendugaan-stok-sumber-daya-ikan-kepada-staf-kkp/show>

⁴ <https://brin.go.id/page/111264/kebijakan-privasi>

⁵ <https://peraturan.bpk.go.id/Details/159344/permen-kkp-no-33permen-kp2019-tahun-2019>

⁶ https://dpmpstsp.jabarprov.go.id/jdih/jdih/detail_jdih/227#:~:text=Peraturan%20Menteri%20Kelautan%20dan%20Perikanan,Pengelolaan%20Perikanan%20Negara%20Republik%20Indonesia

⁷ <https://jdih.maritim.go.id/cfind/source/files/permen-kp/1457d-permen-kp-22-tahun-2021.pdf>



Overview of Knowledge Management



2.1 Defining Knowledge

The terms ‘data’, ‘information’, and ‘knowledge’ are often used interchangeably, yet they possess distinct meanings (Table 1). Each term represents a progression in complexity, where processed data becomes information, and information is transformed into knowledge, with each step adding greater value than the one before it. Hence, beyond managing data, there is a broader concept of KM that needs to be explored.

Table 1. Comparison between data, information, and knowledge, framed in the fisheries context (Sidi et al., 2009; Stenmark, 2002; FAO, 2000)

	Data	Information	Knowledge
Definition	A collection of discrete and objective facts presented as symbols, not yet interpreted and processed, devoid of context	Processed, contextualized, and structured data that contains meaning and answers the questions of who, where, when, and what	A set of framed concepts, methodologies, expert insights, values, and judgments, which answers the questions of why and how
Examples in the fisheries context	● Species	● Population growth rate	● Fisheries theories and concepts
	● Catch, effort, abundance ● Size composition ● Biological (sex, size at maturity, etc.) ● Trip number, fishing gear type, fishing ground coordinate ● Other (ecology, social, economy, etc.)	● Carrying capacity ● Gear selectivity ● Catch per unit effort (CPUE) ● Maximum sustainable yield (MSY) ● Stock size and condition	● Previous studies ● Data collection method(s) ● Stock assessment method(s) ● Exploitation status ● Management goals

	Data	Information	Knowledge
		● Fishing mortality rate ● Total allowable catch ● Spawning area	● Stakeholder roles ● Documents (e.g. books, research articles, management recommendations, plans and measures, fishery laws, regulations, policy) ● Local/indigenous knowledge

2.2 Knowledge Management and Co-production

The continuous updating and evolution of knowledge necessitates effective management. The KM, or ‘knowledge sharing’ is grounded in the creation and dissemination of knowledge (McInerney, 2002). It enables organizations to boost productivity by managing knowledge, making it accessible and useful to individuals and groups (Clarke & Cooper, 2000). The implementation of KM mechanisms has proven to significantly enhance organizational performance quality (Kioko & Muriuki, 2024; Inkinen, 2016). However, it is important to highlight that

most KM theories are constructed with a single organization in mind, and so are their implementations. Whereas KM in a multi-stakeholder context calls for a more robust system, as it operates under a co-production principle.

Knowledge co-production involves contributions from diverse sources of knowledge, viewpoints, and perspectives from various stakeholders in creating knowledge and information to support fisheries management and conservation. Fundamentally, knowledge co-production implies conducting research collaboratively, inclusively, and with full engagement, from identifying research needs, designing studies, collecting data, and interpreting results to apply findings. The goal is to generate knowledge that can benefit all parties involved (Cooke et al., 2020).

Knowledge co-production enhances the relevance and application of scientific knowledge for societal benefits. Collaborative attitudes and active engagement influence the findings produced by fisheries managers. This approach also broadens the perspective on how science is viewed by different stakeholders, allowing for a wider dissemination of knowledge that can be integrated into natural resource governance (Wyborn, 2015). Moreover, interactions with a broad group of stakeholders from various backgrounds can reduce biases in knowledge, making it more creative and objective within a knowledge co-production framework.

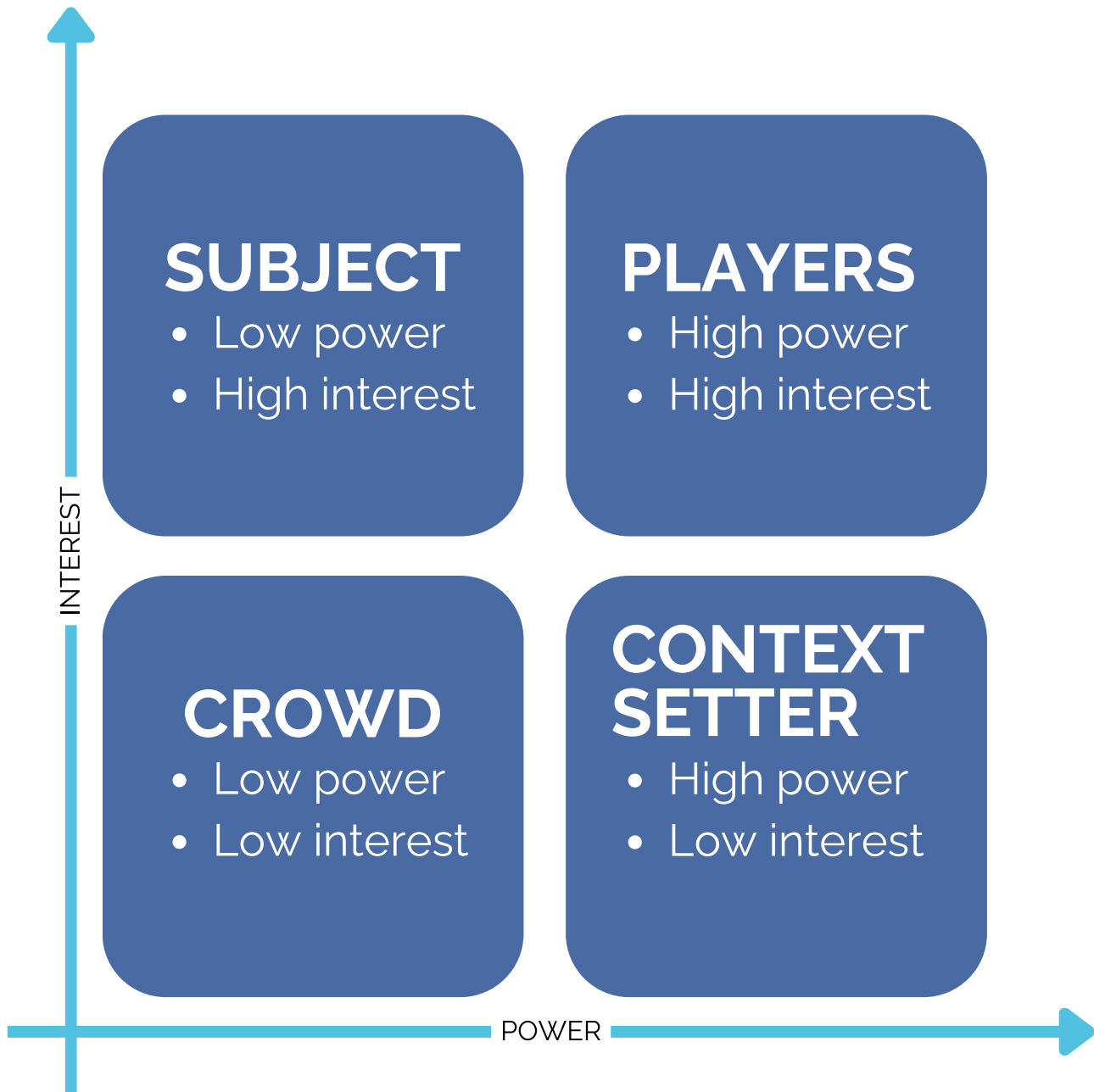
2) (Eden & Ackermann, 1998). Here, “power” is defined as stakeholder influence, while “interest” refers to the stakeholders’ concerns regarding the addressed issue (Guðlaugsson et al., 2020).

2.3 Assessing KM: Stakeholder Mapping

2.3.1 Power-Interest Matrix

The implementation of KM may involve multiple stakeholders with different viewpoints, which necessitates the identification and mapping of relevant stakeholder groups using the power-interest matrix (Ahsan & Pedersen, 2018). This matrix has been employed in various studies to categorize stakeholders into four groups based on their relative power and interest, or their interest and influence concerning the issue at hand (Eden & Ackermann, 1998). Furthermore, understanding the power and interests of different stakeholders can enhance the decision and policy-making process (Guðlaugsson et al., 2020). The stakeholder classification step involves identifying the power and interest of stakeholder groups using a two-dimensional power and interest matrix (Figure

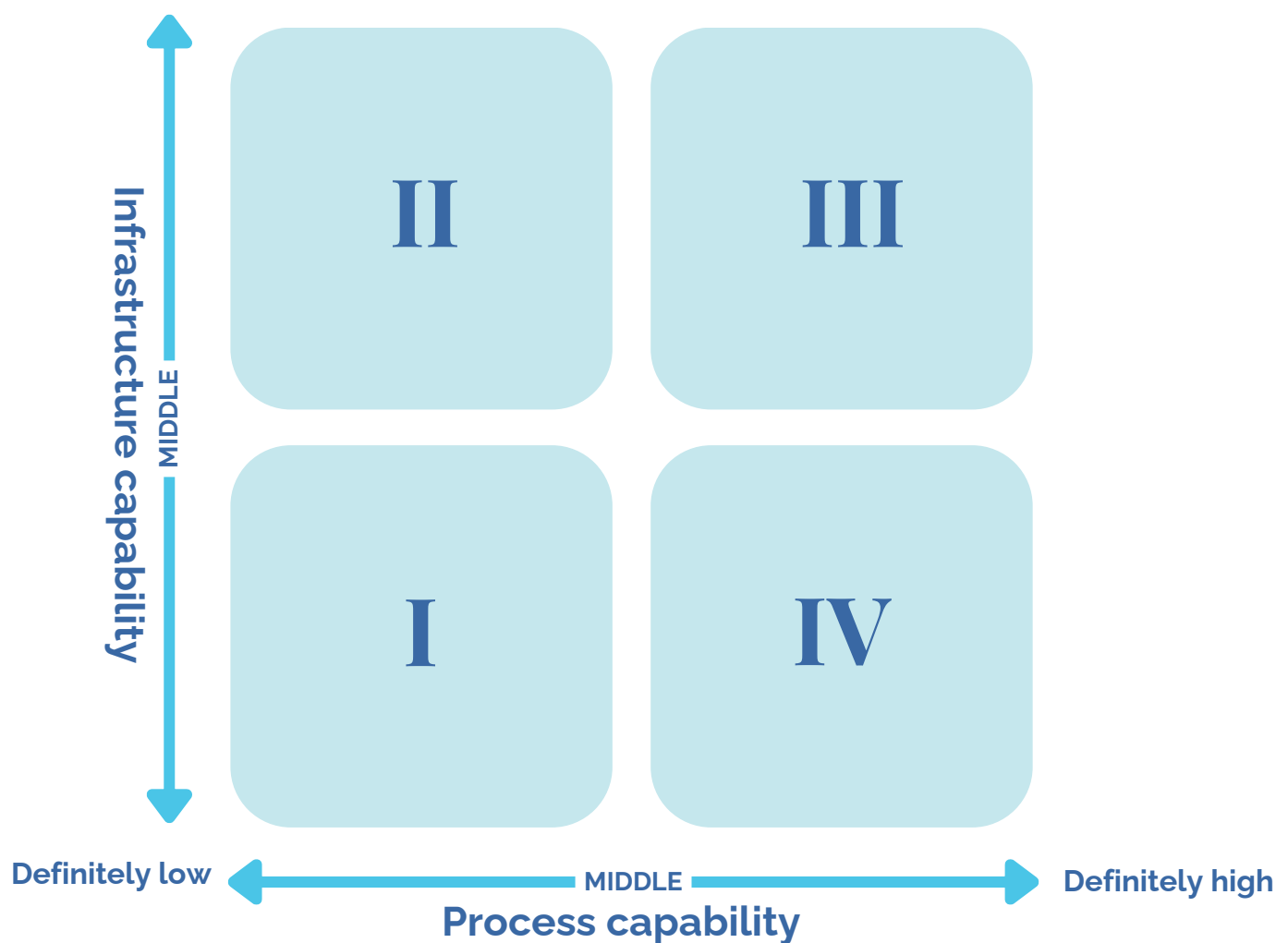
Figure 2. Power and interest matrix (Eden & Ackermann, 1998 in Guðlaugsson et al., 2020)

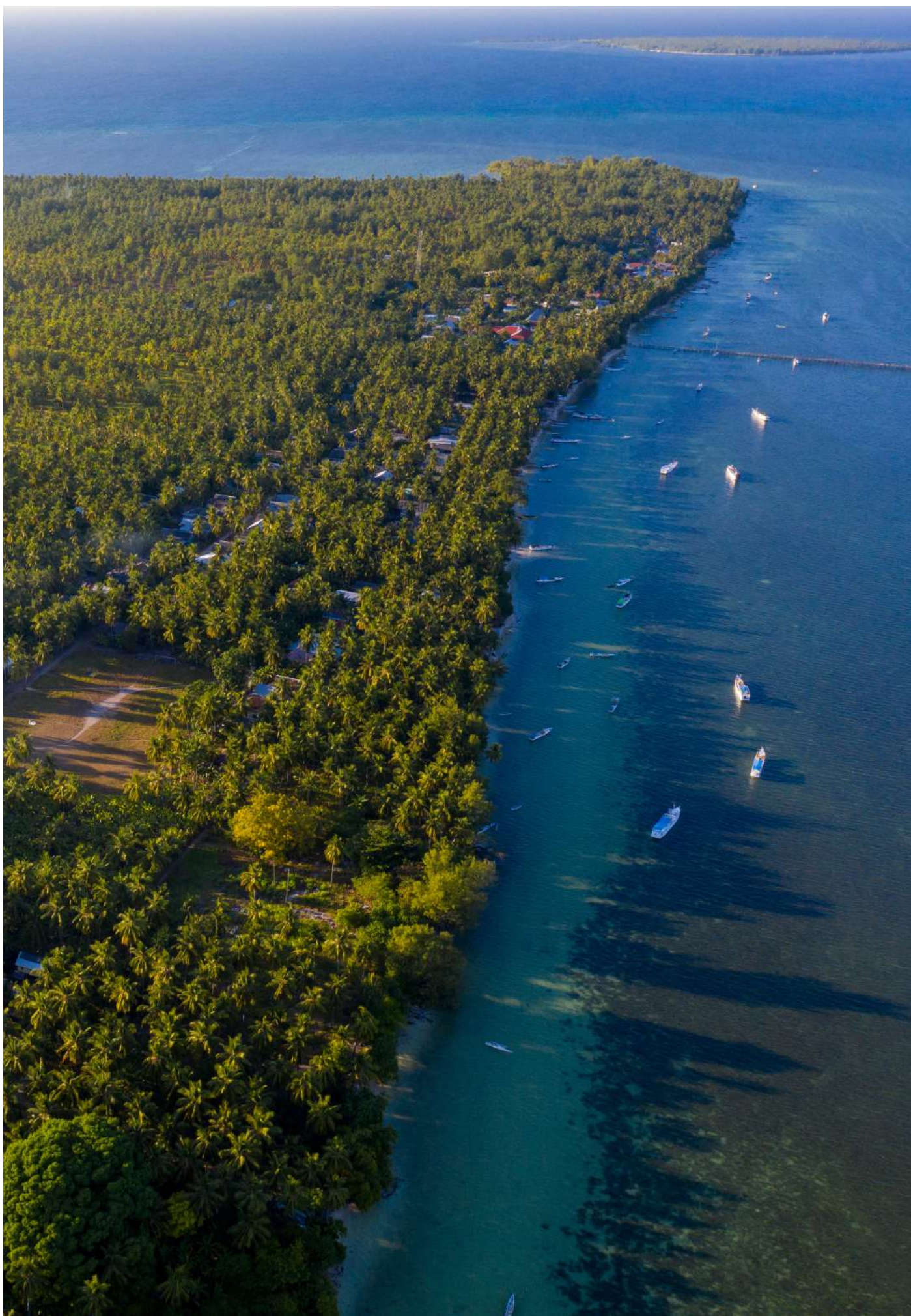


2.3.2 KM Capability

The effectiveness of knowledge sharing among stakeholders hinges on strong KM capabilities (KMC). Organizations gain advantages when knowledge is shared contextually and as needed, through implementing an integrative approach that encompasses all potential knowledge sources and minimizes obstacles to knowledge sharing and organizational learning (Sandhawalia & Dalcher, 2011). A judgment matrix model (Figure 3) with two dimensions is utilized for assessing KMC, namely infrastructure capability (including attributes of technology, structure, and culture) and processes capability (including attributes of acquisition, conversion, application and protection) (Fan et al., 2009).

Figure 3. The judgment matrix model for knowledge management capabilities (Fan et al., 2009)







Situational Analysis



3.1 The Current State of Data and Knowledge Management in Indonesian Fisheries

The fisheries resource management in Indonesia is implemented through a fishing boat license system, an area-based approach and, most recently, a quota-based approach, which are used in combination. To operate in Indonesian waters, a fishing boat and its owner or company must obtain a Fisheries Business License (SIUP) and a Fish Catch Permit (SIPI) or a Fish Transportation Permit (SIKPI), as regulated by the Ministry of Marine Affairs and Fisheries Regulation 58/2020. Small-scale fishers, however, are exempt from permit requirements and instead need only register their vessels, receiving a Fisheries Vessel Registration Certificate (TDKP) as proof of ownership. The area-based approach operates across 11 Fisheries Management Areas of the Republic of Indonesia (WPPNRI), established under the Minister of Agriculture Decree 995/1999, as part of Indonesia's adoption of the FAO's 1995 Code of Conduct for Responsible Fisheries⁸. These 11 areas are designated based on the Indonesian waters' physical, ecological, biogeographic, and oceanographic conditions. Meanwhile, quota-based management is carried out through the Measurable Fishing, as outlined in Government Regulation No. 11/2023 on Measurable Fishing, and the Minister of Marine Affairs and Fisheries Regulation 28/2023 on Measurable Fishing implementation.

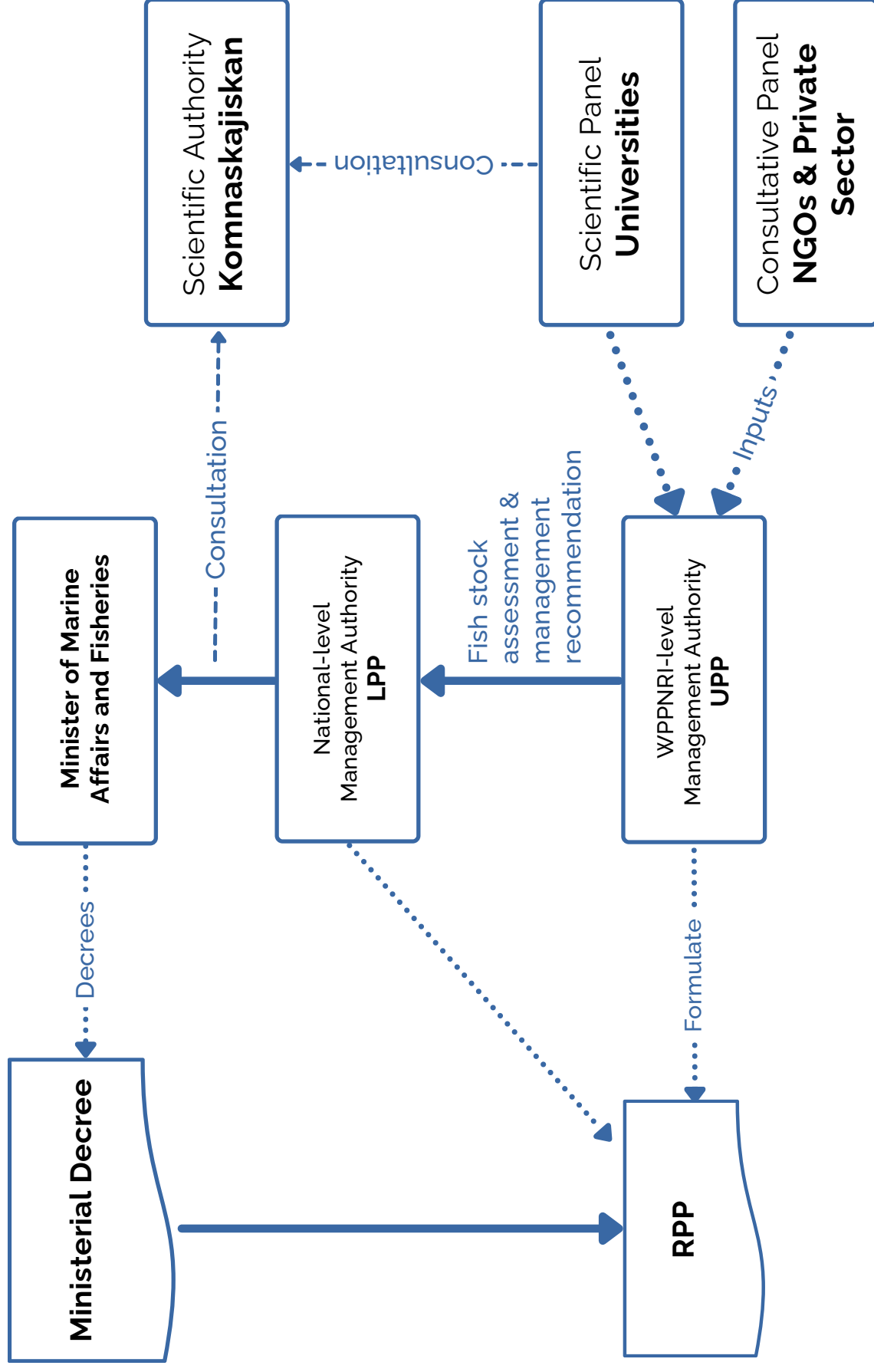
The implementation of fisheries management (Figure 4) is coordinated by the Ministry of Marine Affairs and Fisheries (MMAF), particularly the Fisheries Management Council (LPP) for WPPNRI, led by the Director General of Capture Fisheries of the MMAF. Regulated through Minister of MAF Regulation 33/2019, the LPP is responsible for coordinating and evaluating the implementation of Fisheries Management Plans (RPP), providing advice on their preparation and execution, and offering recommendations to formulate sustainable fisheries management policies in each WPPNRI⁹.

At the national level, the LPP receives data and management recommendations from 11 Fisheries Management Units (UPP). The UPPs are units under LLP that operate at the WPPNRI level. In formulating these recommendations, the UPPs are supported by and receive inputs from scientific panels (i.e. universities and experts) and consultative panels (i.e. NGOs). The LPP head then consults these findings with the National Committee on Fish Stock Assessment (Komnaskajiskan). The Komnaskajiskan is tasked with advising the Minister of Marine Affairs and Fisheries on fish stock estimates, fish resources exploitation rates, and the total allowable catch for each WPPNRI. These values are then formalized annually through a Ministerial Decree, which serve as the basis for quota-based Measured Fisheries Program. The UPP, in turn, formulates regional Fisheries Management Plans (RPP) based on these decrees and the ecological and socio-economic conditions of their respective WPPNRI.

⁸ <https://openknowledge.fao.org/server/api/core/bitstreams/4a456053-db08-4362-875a-2fdc723c1346/content>

⁹ Minister MAF Decree 20/2021 in conjunction with Minister MAF Decree 34/2023





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However, a significant shift has occurred in 2022. The establishment of the National Research and Innovation Agency (BRIN), through Presidential Regulation 74/2019, transferred the research responsibilities and authority from various ministries and agencies to BRIN. This transition included the transfer of researchers including research and technical capabilities from MMAF, such as from MMAF's Fisheries Research Center (Pusat Riset Perikanan / Pusriskan) and Marine Fisheries Research Institute (Badan Riset Perikanan Laut / BRPL), to officially joining BRIN. This shift has led to changes in research functions, data ownership, and the management of research funding. As a result, MMAF no longer has an internal division specifically tasked with conducting research and analyzing fisheries data.

In early 2024, MMAF signed an agreement with BRIN to synergize research and innovation in marine and fisheries sectors¹⁰. Although BRIN is committed to supporting MMAF with science and innovation, there is still no established mechanism for sharing or accessing data and knowledge, resulting in issues such as data gaps and uneven data distribution. The scattered nature of the data also increases the likelihood of overlapping research efforts. Furthermore, adjacent to this issue, fisheries research and knowledge creation at universities tend to operate independently, often without aligning with government needs. Similarly, NGOs tend to follow their own organizational missions and donor requirements, which can lead to a lack of coherence with broader national priorities.

3.2 Fisheries Knowledge Management Stakeholder Mapping

We have conducted a literature review and in-depth interviews with stakeholders to conduct a mapping of the relevant stakeholders (Figure 5 and Figure 6).

Figure 5 illustrates the distribution of stakeholders by both process and infrastructure capabilities, divided into four distinct quadrants. Quadrant I represents fishers and businesses/private sectors, who demonstrate low levels of both process and infrastructure capabilities. Fisher's limited technology use—often restricted to basic mobile phones—and underdeveloped organizational structures contribute to minimal engagement in KM. The business / private sectors are also identified in quadrant I because they lack the infrastructure and capability to process knowledge. Nevertheless, they often collect and analyze their data to maximize profit and improve their business operations (Bradley et al., 2019). Both's focus remains largely operational, with little emphasis on knowledge sharing or management. This lack of involvement in KM highlights the challenges faced by primary resource users in adopting technology and organizational processes for effective knowledge dissemination.

¹⁰Gading, S. (2024). MMAF Signs MoU with BRIN-Barantin to Promote Indonesian Fisheries Ecosystem [KKP Teken MoU dengan BRIN-Barantin buat Dorong Ekosistem Perikanan RI]. Detik Finance. (accessed 29 July 2024). <https://finance.detik.com/berita-ekonomi-bisnis/d-7315594/kkp-teken-mou-dengan-brin-barantin-buat-dorong-ekosistem-perikanan-ri>.

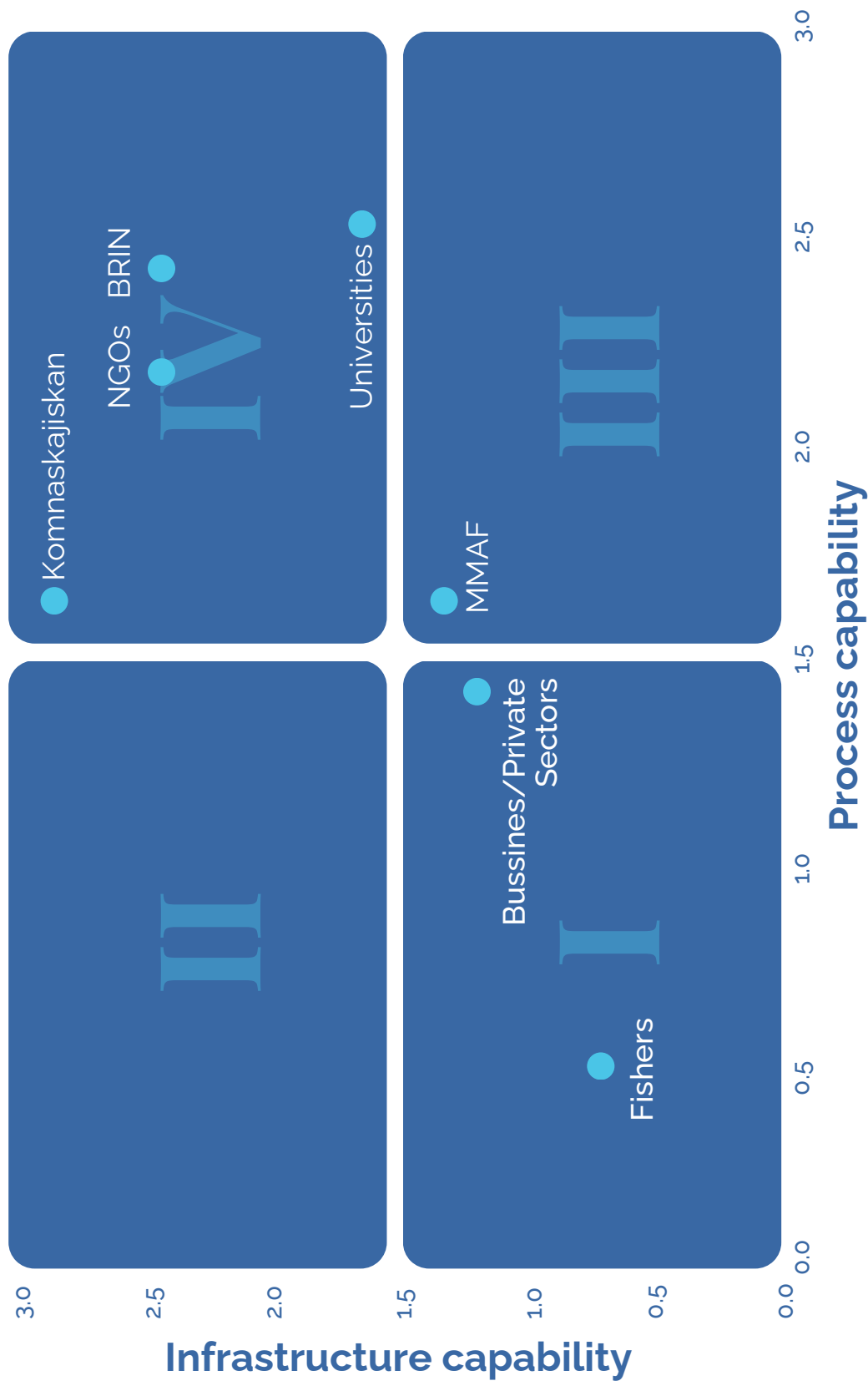


Figure 5. Stakeholder mapping on KM process and infrastructure capability

Quadrant III includes the Marine Fisheries Research Center under the MMAF, which has moderate process capabilities but currently having relatively low infrastructure support. Although the center has a structured approach to data collection, analysis, and dissemination, infrastructure limitations may restrict its overall KM effectiveness. However, its clearly defined roles and responsibilities allow MMAF to contribute significantly to KM through integrated data processes, even if limited by infrastructure¹¹.

Quadrant IV is occupied by high-capability stakeholders, including Komnaskajiskan, the National Research and Innovation Agency (BRIN), non-governmental organizations (NGOs), and universities. These entities possess strong process and infrastructure capabilities, positioning them as leaders in KM and organizational development. BRIN's Center for Data and Information (Pusat Data dan Informasi / Pusdatin), governed by BRIN Regulation 1 / 2021, plays a pivotal role in managing information and technology. Pusdatin's mandate includes preparing and implementing policies in data management, supported by robust technological infrastructure. Komnaskajiskan primarily serves as a quality control body, tasked with reviewing knowledge and developing regulations. However, it does not have the function of directly processing and managing knowledge. Gaps remain in knowledge storage, where universities contribute by providing repositories for printed and digital documents, including theses, research articles, and

publications¹².

Universities also play a crucial role in disseminating research outcomes and integrating KM practices within their organizational structures. Although they generally have effective KM systems, some universities face bureaucratic challenges, such as rigid administrative procedures, which can limit the agility needed for KM decision-making. In contrast, NGOs demonstrate strong involvement in KM with more flexible organizational cultures and administrative structures, enhancing their KM effectiveness. The score attributed to NGOs represents their interest in fisheries data management and reflects the maximum value they can currently achieve. However, both universities and NGOs may encounter funding limitations that hinder their ability to fully implement KM practices.

The distinctions between these quadrants underscore the importance of advancing both process and infrastructure capabilities for effective KM implementation. While Quadrant I stakeholders, such as fishers, are still in the early stages of capability development, Quadrant IV entities demonstrate sophisticated KM capabilities. These advanced stakeholders exert considerable influence on KM practices across sectors, underscoring the need for targeted development strategies to support less-capable groups.

¹¹Minister of Marine Affairs and Fisheries Regulation Number 22 of 2021 concerning the Preparation of Fisheries Management Plans and the Establishment of Fisheries Management Institutions within the Fisheries Management Areas of the Republic of Indonesia

¹²<https://ppid.brin.go.id/posts/lebih-dekat-dengan-infrastruktur-teknologi-informasi>

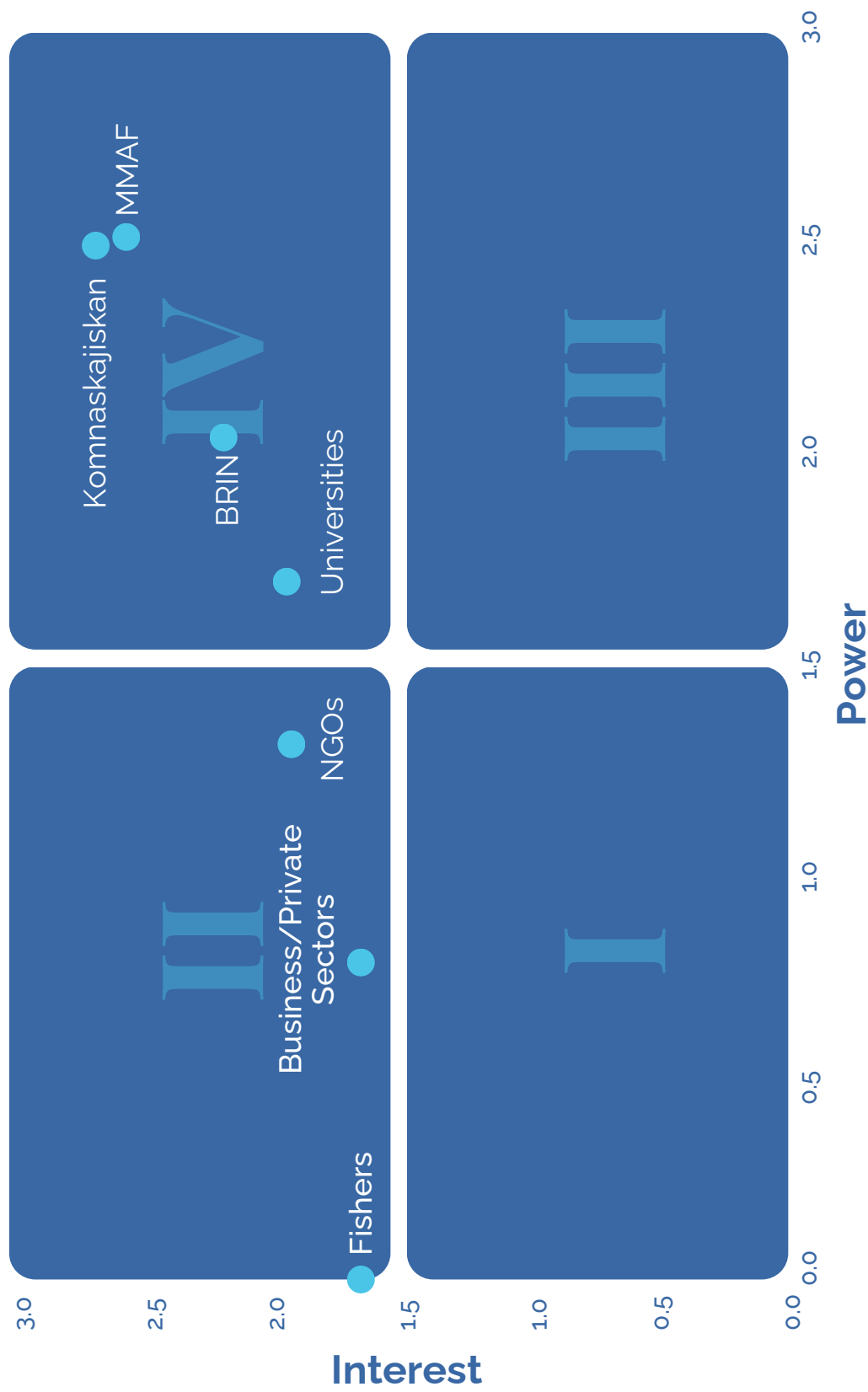


Figure 6. Stakeholder mapping on power and interestThe diagram in Figure 6 maps key stakeholders in fisheries management and KM

processes based on their power and interest. Stakeholders are divided into four quadrants, representing their influence and engagement in KM and policy.

In Quadrant II, fishers, NGOs, and businesses / private sectors are identified. Fishers have a relatively low level of power but a moderate interest in KM due to their reliance on fisheries resources. NGOs also have moderate interest, focusing on advocacy for sustainable practices at the community level. The businesses / private sectors have low power levels but moderate interest in fisheries data to increase business profit (Husein et al., 2015). The business and private sectors in fisheries also engage in collaborative activities with the government to support the blue economy¹³. However, they exhibit limited power to influence policy, with their actions mainly directed at local initiatives (Parikenan et al., 2020).

In Quadrant IV, Komnaskajiskan and the MMAF are positioned as highly interested and powerful actors. These government agencies have significant authority over fisheries policy and KM, shaping the sector through decision-making and implementation. BRIN and universities, also in Quadrant IV but with slightly lower power, contribute to fisheries management through research and knowledge generation^{14, 15}, although their policy influence is somewhat limited compared to Komnaskajiskan and MMAF (Baskoro & Nugroho, 2011).

Komnaskajiskan has the function of validating collected data and creating and establishing regulations based on the collected data.

This stakeholder mapping highlights that Komnaskajiskan and the MMAF hold the greatest influence over fisheries management and KM due to their high levels of interest and power. In contrast, fishers and NGOs, while essential for knowledge dissemination at the grassroots level, lack the authority to drive policy changes. BRIN and universities contribute valuable research and expertise, yet they possess only moderate power compared to the primary policymakers in Quadrant IV.

Despite government efforts to address data gaps and improve data availability through symposiums, stakeholder meetings, and workshops, these initiatives have not yet succeeded in organizing data and knowledge comprehensively. As a result, there remains a critical need for a more integrated flow of data, information, and knowledge to effectively support fisheries management in Indonesia.

¹³<https://indonesia.go.id/kategori/editorial/7880/genjot-produksi-perikanan-via-ekonomi-biru?lang=1> (accessed 10 Oct 2024)

¹⁴<https://www.siej.or.id/index.php/id/ekuatorial/mdpi-dan-brin-perkuat-pengelolaan-perikanan-berkelanjutan-di-indonesia>

¹⁵<https://perikanan.org/public/index.php/id/kegiatan/frci-brin-ipb-university-dan-ykan-beri-pelatihan-pendugaan-stok-sumber-daya-ikan-kepada-staf-kkp/show>

3.3 Ideal Condition: Multi-Stakeholder Fisheries Knowledge Management System

A significant portion of global fisheries, including those in Indonesia, operate under data-limited conditions, which poses challenges for effective management (Costello et al., 2012; Halim et al., 2020). One approach to address this is through the use of a management procedure, often referred to as a ‘harvest strategy.’ This strategy outlines how fishery data are systematically converted through fish stock assessment into management actions, ensuring a more structured and predictable approach to fisheries management (Loneragan et al., 2021).

An integrated digital KM system (Figure 7) can potentially play a crucial role in enhancing data collection, analysis, and policy development. By centralizing data collection within a single platform, data can be standardized across all users, minimizing factors that lead to biases and achieving more reliable data, accurate stock assessments, and estimation approaches (Maunder et al., 2004; Bishop, 2006). The use of cutting-edge technology and sophisticated data systems also enhances data quality by expanding coverage, decreasing time-lags, increasing accuracy, and improving resolution, which supports more adaptive and responsive decisions (Bradley et al., 2019).

A well-structured KM system not only improves

the quality of the produced knowledge but also facilitates its translation, structuring, and interpretation for policy formulation (Lohuizen, 1986), which makes for a potentially better harvest strategy. Additionally, the integration of KM simplifies coordination among stakeholders, including scientific authorities, management authorities, scientific service providers and management partners. This system allows for direct feedback to stakeholders regarding data quality, data needs, knowledge on data collection and analysis methods, knowledge on stock conditions in specific areas, and established management policies.

From a technical standpoint, the integrated platform is designed as an online infrastructure that encompasses everything from database construction and cloud computing to a dashboard that displays the results (Figure 8). This centralized system allows data to be accessible to all registered users within the platform, although a Memorandum of Understanding (MoU) for data sharing is essential to govern access and use. Knowledge protection from illegal and unauthorized use is a crucial aspect of KM, which can employ measures from technological to procedural (Pandey et al., 2018). Given its online nature, the system, particularly the database, must be highly secure to prevent data breaches, which requires robust technology and skilled personnel to manage and protect the platform, such as using firewalls and other security measures. Furthermore, the platform should leverage Application Programming Interfaces (APIs), which function to streamline the process of retrieving and inputting data from and into the database, and also facilitating seamless data sharing across different platforms (Arcuri, 2019).

Utilizing cloud computing for stock assessments allows users to directly access data and conduct analyses through the platform, as cloud-optimized systems offer scalable, efficient, and cost-effective solutions for data processing and storage. It significantly enhances data analysis in fisheries management by enabling real-time data collection and processing (Merrifield et al., 2019), improving data integration and visualization (Vance et al., 2015), increasing transparency (Probst 2020), and fostering collaborative management among stakeholders (Bradley, 2019). More advanced features may include AI-assisted analysis and visualization to automate the monitoring of fish

resources, environmental conditions, and stock assessments (Gladju et al., 2022; Probst, 2020; Ebrahimi et al., 2019).

The results from these analyses, including stock assessments and management recommendations, can be instantly fed back to users and displayed on a publicly accessible dashboard alongside other relevant data. Additionally, the user forum within the platform facilitates identifying data needs, available data sources, ongoing projects, data collection capabilities, and available funding, thereby enhancing collaboration and resource sharing among stakeholders.



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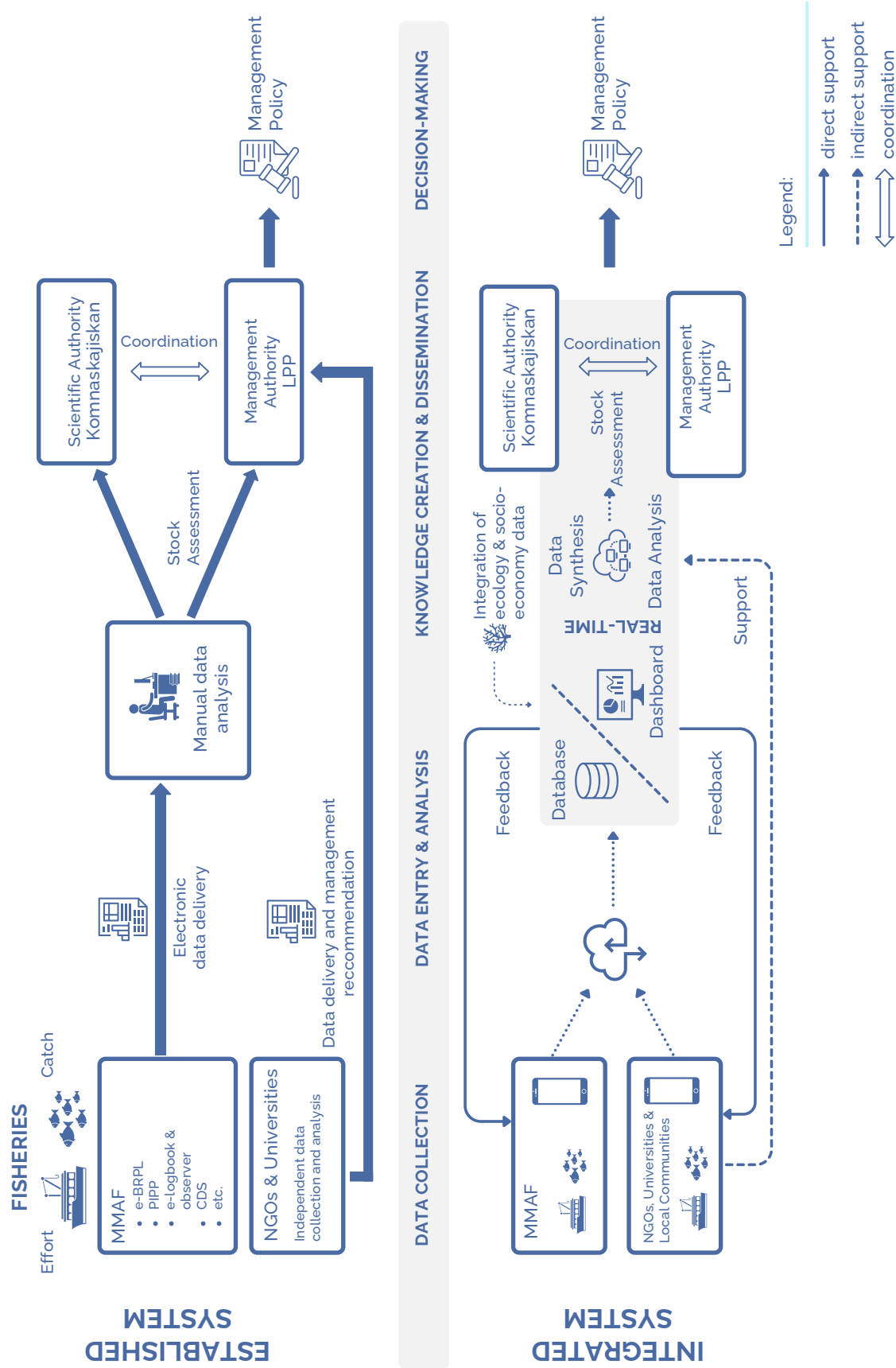


Figure 7. Conceptual diagram of integrated digital data collection and KM system in Indonesia's fisheries (adapted from Bradley et al., 2019). The established system was initiated in ~2019, along with the creation of the LPP. Changes to the established system has been observed since 2022 during the establishment of BRIN. However, currently the established system remains partially implemented despite institutional shifts.

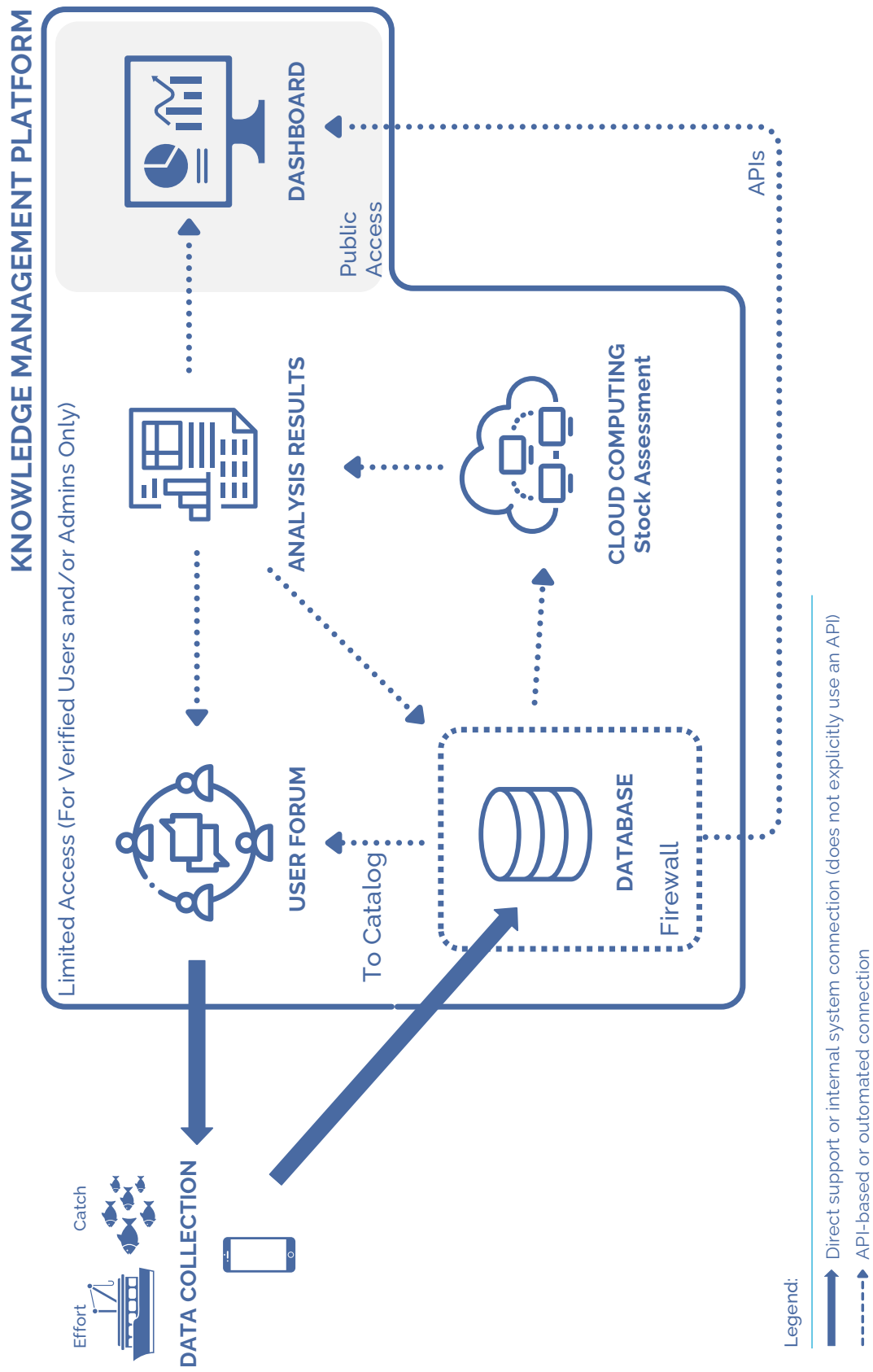


Figure 8. Conceptual diagram of the KM platform infrastructure. ‘Users’ in the user forum refers to registered users. Meanwhile, the dashboard is an open-access platform that can be accessed by everyone outside the data collection scheme, including non-partnered fishers and businesses/private sectors.



4 Recommendations



4.1 First step: Fisheries data cataloguing

A good KM starts with good data management. The fisheries data catalog (Figure 9) represents a critical first step toward establishing an integrated multi-stakeholder KM system. Serving as a centralized repository, the catalog plays a crucial role in managing fisheries data, facilitating the organization of information in a way that enhances the co-production of multi-stakeholder knowledge in Indonesia. Furthermore, the catalog will not only support immediate data management needs but also act as a stepping stone for a seamless transition to a comprehensive KM system.

By streamlining data organization, catalogs make it easier to locate and retrieve information, which is essential for quick and informed decision-making (Ghazaly, 2022). The catalog may also function as centralized repositories for metadata, simplifying the storage and retrieval processes, which are key to effective data management (Subramaniam et al., 2021; Shanmugam & Seshadri, 2016). The catalog should also contain data sharing agreements (DSA) and terms of each data owner and custodian.

The catalog's content is contributed by users and

connected to the databases of the MMAF, enabling stakeholders to identify owners of specific data. This centralized system simplifies the identification of data gaps and overlaps, providing clear insights into data availability. To ensure the catalog is well-maintained, it requires management by an

administrator and regular updates. This necessitates the formation of a working group, potentially involving cross-stakeholder collaboration, to oversee the catalog's management. Furthermore, successful cataloging requires collaboration among all stakeholders, each internally cataloging data within their respective institutions to ensure effective contribution to the collective resource.

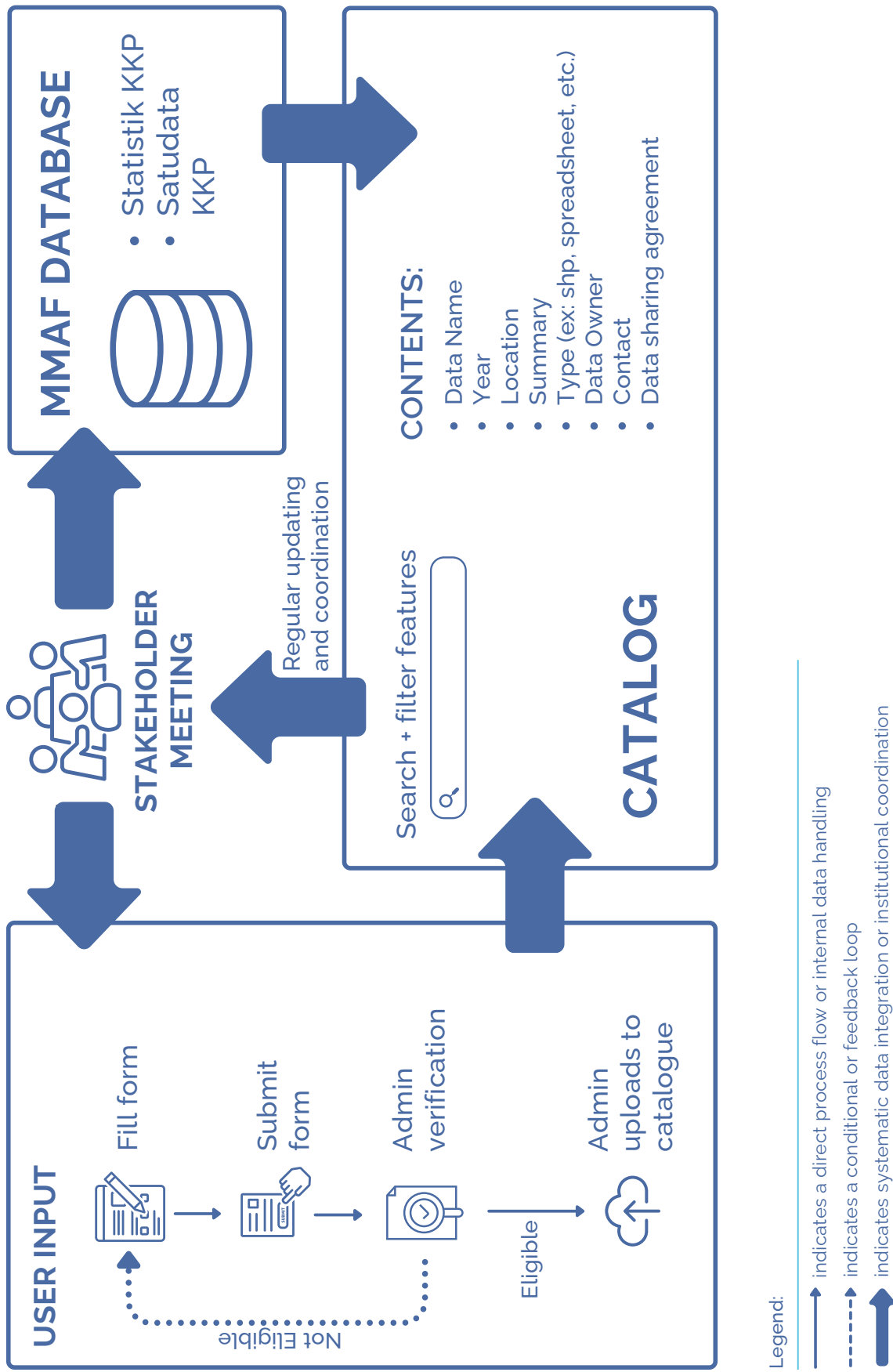


Figure 9. Conceptual diagram of fisheries data catalog

4.2 Toward Effective Fisheries Knowledge Management

In its implementation, the KM cycle (Figure 10) is a continuous process consisting of the creating, sharing, structuring, using, and auditing of knowledge (Sagsan 2006). In this cycle, the results of knowledge auditing feed into the evaluation process, which then forms the basis for subsequent planning and implementation efforts. The KM is best approached from a resource-based perspective, treating knowledge as a valuable resource with a focus on its processing, distribution, and cognition, primarily implemented through information technology (Rowley 2001). However, while technology is key to enabling more effective knowledge creation, sharing, and capture (Sarka et al., 2019), an overreliance on technology at the expense of human factors can lead to the failure of KM initiatives (Malika, 2014).

To mitigate this risk, robust social communication infrastructure is essential, ensuring the involvement and engagement of all stakeholders, which is particularly important in the context of Indonesia's multi-stakeholder fisheries sector. Heaton & Taylor (2002) underline the importance of accounting for the diverse practices, approaches, and needs of KM within different locals and communities of practice. In Indonesia's context, each stakeholder operates within a specific domain, with unique functions and goals, making the knowledge landscape complex,

situational, and context-dependent.

If done well, KM may help bridge these differences by facilitating the alignment of diverse organizational cultures, ultimately standardizing varied practices for cohesive data, information, and knowledge sharing. This standardization would subsequently support the formulation of management policies and ultimately the overarching goal of sustainable management of fisheries resources in Indonesia.

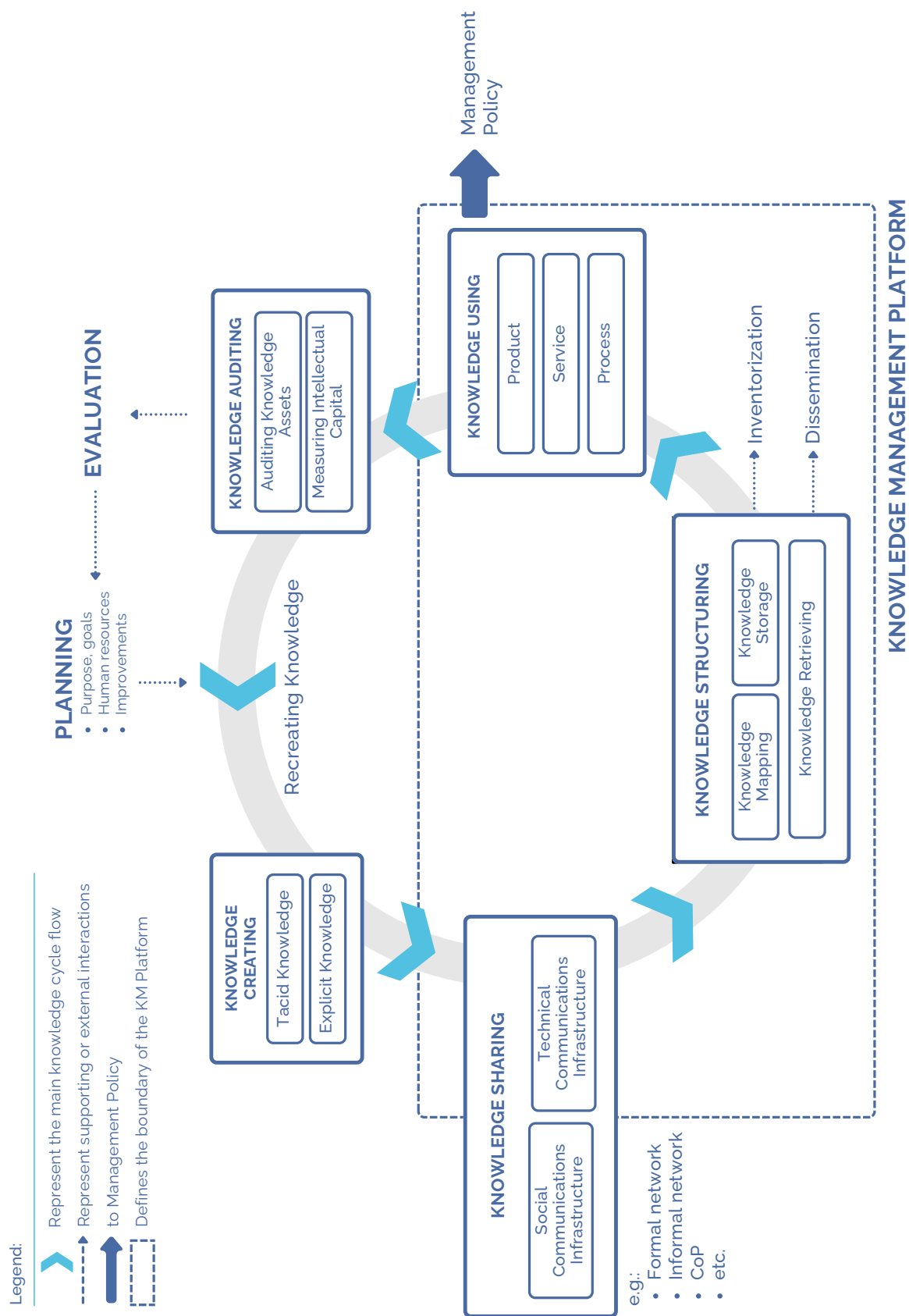


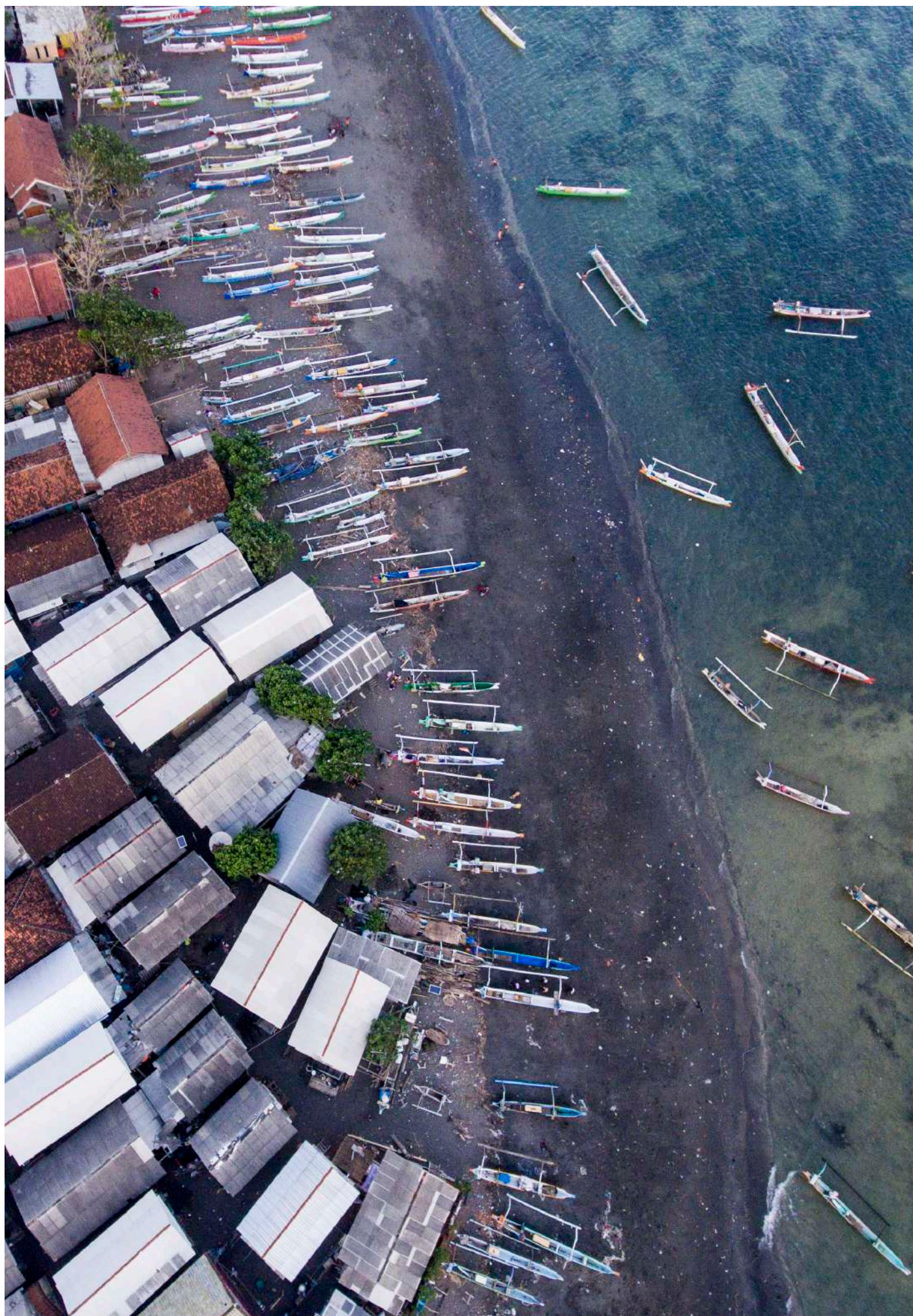
Figure 10. Framework design for the proposed KM cycle in Indonesia's fisheries (adapted from Sagsan, 2006).

4.3 Challenges and Recommendations Matrix

Implementing effective KM in Indonesia's fisheries sector requires addressing the several main challenges. Based on insights from expert interviews with key stakeholders, we identified these challenges, namely limited funding, inadequate technological infrastructure, constrained human resources, and information silos. These challenges, along with targeted recommendations to address them, are outlined in the challenges and recommendations matrix (Table 2).

Table 2. Challenges and recommendations matrix

Challenges	Recommendations	Implementation
Limited Funding	Optimizing other funding sources for KM (collaborating with organizations/universities)	Utilize funding sources outside the government funding which can be allocated for the KM such as funds from private parties/NGOs/donor agencies. Other than funding, the government can utilize in-kind support from stakeholders.
Infrastructure (IT)	Development of capable infrastructure and software	Analyze the technology needs required by KM, comprehensive training in technology, collaboration with technology experts, sharing dataset among stakeholders using APIs
Human Resources	Capacity building to improve human resources competency standards, increasing human resources and awareness of KM knowledge	Outreach (such as FGD, workshop, stakeholder meeting, and training), procurement of specialized functional positions, utilization of in-kind supports
Information silo	Enhancing collaboration with universities, the central government, local governments, and other stakeholders	Develop inter-institutional agreements, in-kind contribution (human resources, services or professional expertise, and knowledge)





Closing Remarks



Effective KM is essential for supporting sustainable fisheries management. While the government bears the primary responsibility, effective management of fisheries and its knowledge is ultimately a shared task, requiring strong collaborative action from all stakeholders. A well-developed KM system not only coordinates efforts but actively facilitates partnerships between diverse actors—from data collection to policy formulation and implementation, ensuring that knowledge is co-produced and widely accessible. This kind of collaborative approach is critical for addressing the complexities of fisheries management and ensuring that diverse perspectives are integrated into the decision-making process.

While challenges like limited funding, inadequate technological infrastructure, constrained human resources, and information silos exist, they can also be addressed through coordinated collaboration involving all relevant stakeholders. Effective KM hinges on the government's ability to leverage the in-kind support, expertise and capacities offered by various stakeholders. Many stakeholders are eager to contribute, and it is crucial that their support is utilized efficiently to build a more integrated and effective management system for Indonesia's fisheries.

AI

Artificial Intelligence

The simulation of human intelligence in machines that are programmed to think, learn, and make decisions, commonly used for data analysis, prediction, and automation.

APIs

Application Programming Interfaces

A set of protocols and tools that allow different software applications to communicate and exchange data with each other.

BRIN

Badan Riset dan Inovasi Nasional/National Research and Innovation Agency

Indonesia's national agency responsible for coordinating, conducting, and advancing research and innovation across various sectors.

BRPL

Badan Riset Perikanan Laut/Marine Fisheries Research Institute

A former Indonesian research institution focused on marine fisheries science and stock assessment.

CDS

Centralized Data System

An integrated system designed to collect, store, manage, and provide access to data from multiple sources.

CoP

Community of Practice

A group of people who share a common interest, profession, or practice and who learn from each other through regular interaction.

DSA

Data Sharing Agreement

A formal agreement that defines the terms, conditions, and responsibilities for sharing data between organizations or stakeholders.

FAO

Food and Agriculture Organization of the United Nations

A specialized UN agency that leads international efforts to defeat hunger and supports sustainable agriculture and fisheries.

FGD

Focus Group Discussion

A qualitative method involving guided group discussions to gather insights, perceptions, and feedback from participants.

HR

Human Resources

The people, skills, and capacities within an organization that support its operations and objectives.

IT

Information Technology

The use of computers, software, networks, and systems to store, process, transmit, and manage information.

KM

Knowledge Management

A systematic approach to capturing, organizing, sharing, and applying knowledge to improve organizational performance and decision-making.

KMC

Knowledge Management Capabilities

The organizational abilities, processes, technologies, and human skills that enable effective knowledge management

Komnaskajiskan

Komisi Nasional Pengkajian Sumber Daya Ikan/National Committee on Fish Stock Assessment

An expert advisory body in Indonesia responsible for providing scientific assessments of fish stock status.

LPP

Lembaga Pengelola Perikanan/Fisheries Management Council

An institutional body responsible for coordinating and advising fisheries management and policy implementation. In Indonesia, LPP operates at WPPNRI level.

MMAF

Ministry of Marine Affairs and Fisheries

The Indonesian government ministry responsible for marine, fisheries, and coastal resource management.

MoU

Memorandum of Understanding

A formal document outlining mutual intentions, roles, and cooperation between two or more parties.

MSY

Maximum Sustainable Yield

The largest long-term average catch that can be taken from a fish stock without compromising its sustainability.

NGO

Non-Governmental Organization

An independent, non-profit organization that operates outside of government, often involved in advocacy, research, or community development.

Pusdatin

Pusat Data dan Informasi/Center for Data and Information

A government unit responsible for managing, integrating, and disseminating data and information.

Pusriskan

Pusat Riset Perikanan/Fisheries Research Center

A research institution under BRIN that conducts scientific studies related to fisheries resources and management.

RPP**Rencana Pengelolaan Perikanan/
Fisheries Management Plan**

A strategic document that outlines objectives, measures, and actions for sustainable fisheries management.

UPP**Unit Pengelola Perikanan/
Fisheries Management Unit**

An operational unit responsible for implementing fisheries management at the regional or local level.

SIKPI**Surat Izin Kapal Pengangkut Ikan/
Fish Transportation Permit**

An official license required for vessels transporting fish and fishery products.

WPPNRI**Wilayah Pengelolaan Perikanan
Negara Republik Indonesia/
Fisheries Management Area of
the Republic of Indonesia**

Officially designated marine areas used as the basis for planning, managing, and regulating fisheries activities in Indonesia.

SIPI**Surat Izin Penangkapan Ikan/Fish
Catch Permit**

An official license authorizing fishing vessels to conduct capture fisheries activities.

SIUP**Surat Izin Usaha Perikanan/
Fisheries Business License**

A permit required to operate a fisheries-related business.

TDKP**Tanda Daftar Kapal Perikanan/
Fisheries Vessel Registration
Certificate**

An official certificate registering a fishing vessel within the national fisheries administration system.

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Author Profile

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Agavia is an Ocean Program officer of the Fisheries Resource Center of Indonesia, Rekam Nusantara Foundation. She is the scientific illustrator and a contributor for 'Kerapu di Indonesia' (Groupers in Indonesia) and 'Kakap di Indonesia' (Snappers in Indonesia) books. She has also contributed to the Indonesia National Chapter of the Fishery Improvement Project COP, particularly in regards to Human Rights and Social Responsibility in Indonesia's small-scale fisheries. Besides fisheries, she is also active in the piloting and implementation of ocean accounts Indonesia, in which she has been central in the development of the Indonesian National Standards for Ocean Accounts, technical modules, and capacity building programs. She has a BS (Hons) in Marine Science and Technology from IPB University, a BA in Sociology from Indonesia Open University, and an LLM from UNSW Sydney.

Nabila Nur Septiani

Nabila holds a bachelor's degree in Resources and Environmental Economics from IPB University. She is a Marine Conservation Economic Specialist at the Fisheries Resource Center of Indonesia, Rekam Nusantara Foundation. She is actively involved in the preparation for the establishment and expansion of Indonesia's National Marine Protected Areas through cost-benefit analysis (CBA). At the local level, she has also contributed to the CBA analysis of the Liukang Tangaya Marine Protected Area. In addition to CBA, she is currently contributing to the pilot and implementation of Ocean Accounts in Indonesia, as well as to the development of technical guidelines for Ocean Accounts and related capacity-building programs.

Marsha Hamidah

Marsha Hamidah was a GIS Staff member at the Fisheries Resource Centre of Indonesia, Rekam Nusantara Foundation. She held a bachelor's degree in Marine Science and Technology from IPB University. Her professional expertise lay in spatial and temporal data digitization, remote sensing, and geospatial analysis to support marine and coastal resource management. At FRCI, Marsha was responsible for operating GIS and remote sensing software, particularly ArcGIS, to analyze mangrove, seagrass, and coral reef ecosystems. She was actively involved in satellite imagery digitization and spatial data analysis for the development and implementation of Ocean Accounting in Indonesia. Her work included data collection, processing, and analysis to support evidence-based marine conservation planning. She is currently working at the Geospatial Information Agency (BIG) as a surveyor.

Prof. Dr. Budy Wiryawan

Budy is a Senior Lecturer at Bogor Agricultural University. He was a head of Department for Marine Fisheries in the Faculty of Fisheries and Marine Sciences from 2009-2017. He has been certified as EIA Compiler and Coastal Management Expert by Indonesia Coastal Management Expert Association/HAPPI). Before, he experienced for numbers of large scale of fisheries, coastal management and marine conservation programs, including the ADB funded COREMAP Phase II Project (2005-2007) as Marine Management Area and Environmental Specialist. He appointed as Marine Program Manager for The Nature Conservancy in East Kalimantan (2003-2004), and USAID funded Coastal resource Management Project in in East Kalimantan (2004-

2005) as Technical Advisor. He was an advisor for Berau Regency Marine Protected Area Development (2003-2005). He was teaming a DHI's Ecological Monitoring for PT Newmont Nusa Tenggara in 2009-2010. He was a member for Jakarta Bay Environmental Assessment, and some of Strategic Environmental Assessment (SEA) funded by Danish Embassy and Ministry of Environment and Forestry during 2010-2012. He appointed also an Evaluator for Impact Evaluation of Marine Resource Projects funded by USAID in Indonesia in 2014, while in 2015 he appointed by World Bank as an Evaluator for COREMAP II project implementation. Until February 2018, he led some Environmental Strategic Impact Assessment capacity building project funded by DANIDA for Local Government Planning (SEA of Regional Development Plan and Spatial Planning) for some Regencies and drafting Ministry Regulation for Spatial Planning SEA by Ministry Agrarian and Spatial Planning. His professional experience lies predominantly with coastal resource management, capture fisheries and conservation, and he's had extensive experience in planning, managing and executing marine research programs.

Dr. Arisetiarso Soemodinoto

Ari is a seasoned marine conservationist with more than 30 years of experience, who has worked on conservation, environmental, and community development. Ari has strong expertise in ecology, marine conservation planning, and the evaluation of management effectiveness and conservation impacts. Ari holds a PhD from Monash University and two master's degrees from the National University of Singapore and the Bandung Institute of Technology. Ari has extensive experience facilitating the development of conservation areas in Indonesia, as well as community-based tourism and fisheries

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Dr. Toni Ruchimat

Toni has almost 40 years working experiences managing and oversee projects/ programs, varied from research, community based natural resources management, marine protected area management and network, species conservation, and sustainable fisheries management. Long term experiences working within the Government of Indonesia and has strong network with policy and decision makers, researchers, NGO's, academicians, coastal and fishermen communities at national and international. Various experiences as expert and team leader for fisheries management plans, EAFM indicators, MPA management effectiveness monitoring tool, coral reef management, Various experiences leading Indonesia delegation in international meetings; CTI, SSME, IOTC, SEAFDEC, CCSBT, WCPFC, IORA, BOBLME, ATSEA, HLP, COFI-FAO, UNFSA-UNEA. Long term experiences in various research and field work activities with numbers of publications varied from technical report, books, proceeding, papers both national and international levels.

Annisya Rosdiana

With a background in marine science from IPB University, Annisya has over 10 years of experience in advancing sustainable ocean development in Indonesia. Her work spans from local sites to multinational initiatives, bridging on-the-ground action with global collaboration. She has managed various projects to support conservation and biodiversity efforts, strengthening the impact of government programs, multilaterals, philanthropies, south-to-south cooperation, and development partners. Deeply passionate about the ocean, she dedicated her expertise, along with the Fisheries Resource Center of Indonesia, Rekam Nusantara Foundation.

Dr. Heidi Retnoningtyas

Heidi holds a master's degree in marine science from IPB University and a doctorate in Agriculture and Environmental Studies from the University of Rostock, Germany, specializing in marine fish parasites as proxies for stock discrimination. She did her tenure as Fisheries Coordinator at the WCS Indonesia Program and worked closely with stakeholders at the local and national level, focusing on developing fisheries management plans and harvest strategy. She started her career as a research and teaching assistant at the IPB University, reflecting her commitment to disseminating knowledge—a commitment that she has carried into her transformative work in non-governmental organizations. She moved from academia to the dynamic world of NGOs with a desire for flexibility and a drive to engage with a broader spectrum of society. In her current role as the Director of the Ocean Program, she steers the ship at the Fisheries Resource Center of Indonesia (FRCI), providing an

alternative to fisheries analysis and management of sustainable marine resources based on scientific data. The core programs of FRCI under her leadership are ocean accounting, marine protected areas, shark and ray conservation, and sustainable fisheries management.

Dr. Irfan Yulianto

Irfan has more than 20 years working experiences, varied from community based natural resources management, marine protected area management and network, species conservation, and sustainable fisheries. Long term experiences working closely with coastal communities, district and provincial governments, high level decision and policy makers at Ministry of Marine Affairs and Fisheries. Strong networking along with national and international NGOs, researchers, academicians, decision makers and policy makers. Strong ability related database management, GIS and Marxan analysis, and RStudio for fisheries sciences.







